

# Piecewise-Affine Local Volatility

Calibrating a Dupire surface from sparse quotes: model, PDE, gradients, and speed

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## Abstract

This note documents the Vol-Fitter’s full local-volatility model: a continuous piecewise-affine ( $P_1$ ) local-variance surface  $\nu_\theta(t, x)$ , calibrated directly to sparse vanilla and variance-swap quotes through the forward Dupire PDE as an implicit pricing map. Unlike the slice models (Notes 01–03) this is a genuine two-dimensional surface: strict positivity of the nodal variances makes the implied prices butterfly- and calendar-clean up to scheme noise, which the model *measures* rather than assumes. We derive the normalized forward Dupire equation and prove its arbitrage-propagation property, the  $P_1$  parametrization and its positivity guarantee, and — the mathematical heart of this revision — the discrete theory: the implicit-Euler step matrix is an  $M$ -matrix, giving a discrete maximum principle, unconditional  $\ell^\infty$  stability and a pivot-free Thomas march, while Crank–Nicolson provably loses monotonicity on coarse grids (and was observed to manufacture an arbitrage). We derive the discrete tangent and adjoint sensitivities and show both are what production actually runs: the matrix-free Gauss–Newton solver touches the Jacobian only through the tangent product  $Jv$  and the adjoint product  $J^\top w$ . The engineering story — a compiled vectorized-Thomas march ( $\sim 6.5\times$ ), a stall-based early-stop, warm seeds, and the matrix-free GN correctly scoped to where the dense SVD actually dominates — cut the cold fit  $\sim 3\text{--}6\times$ , with recalibrations near-instant. Two case files close the loop: the six-day weekly that broke the strike grid ( $108 \rightarrow 23.5$  vol bp), and the convex-wing constraint that flattened SPY until it was confined to the extrapolation tail ( $25.7 \rightarrow 2.6$  vol bp). A synthetic round trip recovers a known surface to 0.6 vol bps; the production fit on the static Bloomberg fixture reaches 2.8 vol bps on SPY and 11.7 on NVDA.

## Contents

|          |                                                             |          |
|----------|-------------------------------------------------------------|----------|
| <b>1</b> | <b>The production problem</b>                               | <b>3</b> |
| <b>2</b> | <b>The forward Dupire equation</b>                          | <b>3</b> |
| <b>3</b> | <b>The piecewise-affine surface</b>                         | <b>4</b> |
| 3.1      | Parametrization . . . . .                                   | 4        |
| 3.2      | Grid design . . . . .                                       | 4        |
| <b>4</b> | <b>The discrete pricing map</b>                             | <b>5</b> |
| 4.1      | Discretization . . . . .                                    | 5        |
| 4.2      | Why not Crank–Nicolson . . . . .                            | 6        |
| 4.3      | Observation operator and no-arbitrage diagnostics . . . . . | 6        |

|           |                                                    |           |
|-----------|----------------------------------------------------|-----------|
| <b>5</b>  | <b>Variance swaps</b>                              | <b>6</b>  |
| <b>6</b>  | <b>The calibration objective</b>                   | <b>7</b>  |
| <b>7</b>  | <b>Sensitivities: tangent and adjoint</b>          | <b>7</b>  |
| <b>8</b>  | <b>Solvers and the speed story</b>                 | <b>8</b>  |
| 8.1       | Where the time goes . . . . .                      | 8         |
| 8.2       | What worked . . . . .                              | 9         |
| <b>9</b>  | <b>Case files</b>                                  | <b>9</b>  |
| <b>10</b> | <b>Worked examples</b>                             | <b>11</b> |
| 10.1      | Synthetic round trip . . . . .                     | 11        |
| 10.2      | The Bloomberg benchmark . . . . .                  | 11        |
| <b>11</b> | <b>What is original, and limitations</b>           | <b>12</b> |
| <b>12</b> | <b>Traceability</b>                                | <b>12</b> |
| <b>A</b>  | <b>Hyperparameter atlas</b>                        | <b>14</b> |
| <b>B</b>  | <b>Performance notes</b>                           | <b>15</b> |
| <b>C</b>  | <b>Reference implementation: Dupire extraction</b> | <b>16</b> |

# 1 The production problem

A slice model fits one expiry at a time and stitches expiries together with a calendar constraint. Production needs the opposite object too: one surface that prices *every* strike and expiry of a name simultaneously — for scenario transport (Note 12), variance swaps (Note 08), and as the projection target of the graph extrapolator (Note 14). The classical route — fit an implied surface, then *differentiate* it through Dupire’s formula — is numerically explosive exactly where production data lives: sparse wings, short tenors, stale quotes. Dividing two noisy numerical derivatives manufactures negative variances and spikes that no downstream PDE survives.

## Invariant

1. Every optimizer iterate is a genuine, strictly positive local-variance surface — positivity is carried by the *parameters*, never policed by a penalty.
2. The pricing map never differentiates market data: quotes enter only as targets of a forward PDE solve.
3. The discrete prices obey a maximum principle — the scheme cannot manufacture negative densities; the residual scheme noise is *measured* by explicit diagnostics, not assumed away.
4. Every short-dated rescue and speed lever is gated so that surfaces which do not need it are byte-identical.

The Vol-Fitter therefore follows the Andreasen–Huge philosophy [2]: *parametrize* the local variance directly, price *forward* through the Dupire PDE, and calibrate the parameters to the quotes.

## Heuristic

Extracting local vol by  $\sigma_{\text{loc}}^2 = \partial_T C / (\frac{1}{2} K^2 \partial_{KK} C)$  divides two noisy derivatives of a sparse surface — garbage in the wings. Calibrating *forward* never differentiates the data: it asks “what positive surface, run through the PDE, best reprices the quotes?” Positivity is imposed on the parameters, so every iterate is arbitrage-clean by construction — the LQD philosophy of Note 01, one dimension up.

# 2 The forward Dupire equation

**Assumption 1.** Work in forward-normalized units:  $x = K/F_T$  is the normalized strike and  $c(T, x)$  the normalized call, so deterministic rates, borrow and dividends are absorbed into the forward (Note 06). The maturity variable is the *event-weighted* variance time  $\tau$  of Note 11 — the diffusion clock — and the quotes are de-Americanized mids (Note 05). The model is a pure diffusion: no jumps, so a genuine short-dated event spike is only approximated.

Writing local variance  $\nu(t, x) = \sigma_{\text{loc}}^2(t, x)$ , the forward Dupire equation is

$$\partial_T c(T, x) = \frac{1}{2} \nu(T, x) x^2 \partial_{xx} c(T, x), \quad c(0, x) = (1 - x)^+.$$
 (1)

It is a forward parabolic PDE in  $(T, x)$  with the payoff as initial data [1]. Its structural virtue is that positivity of  $\nu$  *propagates* both static-arbitrage conditions:

**Proposition 1** (Positive local variance propagates no-arbitrage). *Let  $c$  solve (1) with  $\nu \geq 0$  bounded and the payoff initial condition. Then for all  $T > 0$ : (i)  $\partial_{xx}c(T, \cdot) \geq 0$  (no butterfly arbitrage); (ii)  $\partial_T c(T, x) \geq 0$  (no calendar arbitrage at fixed normalized strike).*

*Proof sketch.* Let  $p = \partial_{xx}c$ . Differentiating (1) twice in  $x$  gives the forward (Fokker–Planck-type) equation  $\partial_T p = \frac{1}{2} \partial_{xx}(\nu x^2 p)$ , whose initial datum is the Dirac mass of the payoff kink at  $x = 1$  — nonnegative. Such equations preserve nonnegativity (parabolic maximum principle applied to the adjoint, or probabilistically:  $p(T, \cdot)$  is the density of the diffusion  $dX_t = \sqrt{\nu(t, X_t)} X_t dW_t$  started at 1). With  $p \geq 0$  established, (ii) is immediate from (1):  $\partial_T c = \frac{1}{2} \nu x^2 p \geq 0$ .  $\square$

The whole design of this model is to make proposition 1 survive discretization (section 4) and parametrization (section 3) — and then to *measure* what little the discretization leaks (section 4.3).

## 3 The piecewise-affine surface

### 3.1 Parametrization

**Definition 1** ( $P_1$  local-variance surface). On a tensor grid of  $n_t \times n_x$  vertices  $(\tau_i, \xi_j)$  the local variance is the continuous piecewise-affine interpolant

$$\nu_\theta(t, x) = \sum_{i,j} \theta_{ij} \phi_{ij}(t, x), \quad (2)$$

where the hat functions  $\phi_{ij}$  are barycentric on a (cached Delaunay) triangulation of the vertices — genuinely affine on each triangle — and  $\theta_{ij} = \nu(\tau_i, \xi_j) > 0$  are the *nodal local variances*, the calibration parameters. The flat vector is time-major  $\theta = \text{vec}(\theta)$ .

(A bilinear tensor interpolant is kept for comparison, but bilinear is not affine on cells; the Delaunay barycentric basis is the golden convention.)

**Proposition 2** (Nodal positivity is surface positivity). *On each triangle a  $P_1$  function is a convex combination of its three vertex values. Hence  $\theta_{ij} \geq \nu_o > 0$  for all  $i, j$  implies  $\nu_\theta(t, x) \geq \nu_o$  everywhere on the grid’s hull, and the basis is a partition of unity (test-locked).*

This is why the calibration needs only *box bounds* on  $\theta$ : positivity of a two-dimensional surface reduces to positivity of finitely many numbers. In production the box is  $[\text{varLo}, \nu_{\text{hi}}]$  with  $\text{varLo} = 0.0025$  (local vol 5%) and an *adaptive* cap  $\nu_{\text{hi}} = \min(\max(0.36, (\text{lvVolCapMult} \cdot \sigma_{\text{max}})^2), 16)$  — at least 60% vol, three times the largest observed implied vol if that is larger, hard-capped at 400%. (The tighter  $[0.005, 0.20]$  box quoted in older revisions is only the model-layer function default used by the golden tests.)

Beyond the lowest strike vertex the surface extrapolates local variance linearly with slope `left_extrap_a` times the first interior cell’s slope —  $a = 0$ , the hot-path default, is a flat clamp; a free  $a$  is fitted only when a var-swap quote identifies the deep-put tail (Note 08). The right wing is flat-clamped.

### 3.2 Grid design

The strike vertices are delta-spaced (at the deltas  $\{0.01, 0.02, 0.05, 0.10, 0.25, 0.40, 0.50\}$  and their call mirrors, sized by the *longest* expiry’s ATM std) and floored per expiry so a narrow short-dated

smile still lands enough vertices on its curvature (`gridXNodes`, `gridXMinPerExpiry`; the case file of section 9 is exactly the failure that motivated the floor). The time vertices (`gridTNodes`) carry the term structure. The grid is deliberately modest — a dozen strike vertices, ten in time — because each vertex is a free parameter and PDE cost scales with the grid; the roughness regularizer (section 6) supplies the smoothness a coarse grid would otherwise lack.

## 4 The discrete pricing map

### 4.1 Discretization

The forward map (1) is solved on a uniform strike grid with step  $\Delta x = 0.01$  on  $x \in [0, x_{\max}]$ ,  $x_{\max} \gtrsim 2.5$  ( $\sim 250$  nodes; the short-dated refinement of section 9 can tighten  $\Delta x$  to  $1/400$ ), with Dirichlet boundaries  $c(\cdot, 0) = 1$ ,  $c(\cdot, x_{\max}) = 0$ , by fully implicit (backward) Euler steps of size  $\Delta T = 0.01$ , the local variance entering at the *new* time level. On a (possibly non-uniform) grid with gaps  $h_i^- = x_i - x_{i-1}$ ,  $h_i^+ = x_{i+1} - x_i$ , the operator  $A$  encoding  $\frac{1}{2}\nu x^2 \partial_{xx}$  has the three-point stencil

$$A_{i,i\mp 1} = \nu_i a_i^\mp, \quad A_{ii} = -\nu_i(a_i^- + a_i^+), \quad a_i^\mp = \frac{x_i^2}{(h_i^- + h_i^+) h_i^\mp} > 0, \quad (3)$$

and each step solves the tridiagonal system

$$B^{n+1}U^{n+1} = U^n + (\text{boundary terms}), \quad B^{n+1} = I - \Delta T A^{n+1}. \quad (4)$$

**Proposition 3** (The implicit step matrix is an  $M$ -matrix). *For any  $\nu \geq 0$  and any  $\Delta T > 0$ , the matrix  $B = I - \Delta T A$  of (4) satisfies: positive diagonal  $B_{ii} = 1 + \Delta T \nu_i(a_i^- + a_i^+) \geq 1$ , non-positive off-diagonals  $B_{i,i\mp 1} = -\Delta T \nu_i a_i^\mp \leq 0$ , and unit row sums at interior rows. Hence  $B$  is a strictly diagonally dominant nonsingular  $M$ -matrix, and  $B^{-1} \geq 0$  elementwise.*

*Proof.* The sign pattern and row sums are read off (3):  $B_{ii} - |B_{i,i-1}| - |B_{i,i+1}| = 1 + \Delta T \nu_i(a_i^- + a_i^+) - \Delta T \nu_i(a_i^- + a_i^+) = 1 > 0$ , so  $B$  is strictly diagonally dominant with positive diagonal and non-positive off-diagonals — the defining pattern of a nonsingular  $M$ -matrix, and  $M$ -matrices are inverse-positive:  $B^{-1} \geq 0$  (see e.g. Berman–Plemmons [5], Theorem 6.2.3; concretely, Jacobi iteration for  $Bu = v$  with  $v \geq 0$  has nonnegative iterates and converges by dominance).  $\square$

**Corollary 1** (Discrete maximum principle and unconditional stability). *For any  $\Delta T > 0$  the scheme (4) is monotone ( $U^n \geq V^n$  with the same boundary data implies  $U^{n+1} \geq V^{n+1}$ ) and satisfies  $\min(\min_i U_i^n, c_{\text{bdry}}) \leq U_i^{n+1} \leq \max(\max_i U_i^n, c_{\text{bdry}})$ : the implicit march is unconditionally  $\ell^\infty$ -stable, with no CFL restriction, and cannot create new extrema — in particular it cannot manufacture the oscillations that read as negative densities.*

*Proof.* Monotonicity is  $B^{-1} \geq 0$  applied to  $U^n - V^n \geq 0$ . For the bounds, let  $M$  be the larger of  $\max_i U_i^n$  and  $c_{\text{bdry}}$ : by the unit row sums  $B(M\mathbf{1}) \geq M\mathbf{1}$  componentwise, and  $M\mathbf{1}$  dominates the right-hand side of (4), so  $U^{n+1} \leq M\mathbf{1}$  by inverse positivity; the lower bound is symmetric.  $\square$

Strict diagonal dominance is also why no pivoting is ever needed: each step is a no-pivot Thomas march — the kernel the compiled march of appendix B vectorizes. (The scalar listing below is the *algorithm*; the production kernel `affine_march._march` is its multi-RHS vectorization with the sensitivity columns as the contiguous SIMD inner loop.)

Listing 1: The no-pivot tridiagonal (Thomas) solve for one implicit step (4) — the scalar algorithm that `models/localvol/affine_march.py` vectorizes; matches a dense solve to  $10^{-10}$ .

```

1  def thomas(a, b, c, d):          # a: sub-, b: main, c: super-diagonal
2      n = len(d); cp = np.empty(n); dp = np.empty(n)
3      cp[0], dp[0] = c[0]/b[0], d[0]/b[0]
4      for i in range(1, n):        # forward sweep (no pivoting: Prop. M-
          matrix)
5          m = b[i] - a[i]*cp[i-1]
6          cp[i], dp[i] = c[i]/m, (d[i] - a[i]*dp[i-1])/m
7      x = np.empty(n); x[-1] = dp[-1]
8      for i in range(n-2, -1, -1):
9          x[i] = dp[i] - cp[i]*x[i+1]
10     return x

```

## 4.2 Why not Crank–Nicolson

**Remark 1** (CN loses monotonicity exactly when it matters). Crank–Nicolson replaces (4) by  $(I - \frac{\Delta T}{2}A)U^{n+1} = (I + \frac{\Delta T}{2}A)U^n$ . The implicit factor is still an  $M$ -matrix, but the *explicit* factor has diagonal  $1 - \frac{\Delta T}{2}\nu_i(a_i^- + a_i^+)$ , which is nonnegative only under the CFL-like bound  $\Delta T \leq 2/(\nu_i(a_i^- + a_i^+))$  — violated precisely on coarse strike grids with high local variance. When it is violated the scheme is second-order accurate but *not* monotone, and the payoff kink excites the classic CN oscillations. This is not hypothetical: the opt-in Rannacher/CN scheme (two implicit start-up steps damping the kink [4], then CN) bought only  $\sim 1.1\times$  and produced a measured coarse- $x$  arbitrage violation on a test surface — so the default stays fully implicit, trading one order of time accuracy for corollary 1.

## 4.3 Observation operator and no-arbitrage diagnostics

The observation operator reads the normalized call at each quote’s  $(T, x)$  off the marched solution by interpolation. Because the discrete surface is arbitrage-clean only up to scheme noise, the model *measures* the residual: a butterfly tolerance on  $\partial_{xx}c$  and a calendar tolerance on  $\partial_T c$  (both  $10^{-8}$ ), surfaced as diagnostics and gating rather than assumed. A separate, finer log-strike Crank–Nicolson/Rannacher validator (`models/localvol/pde.py`; `atlas`) exists purely to check the production march against an independent discretization.

## 5 Variance swaps

A variance-swap quote is the fair strike of realized variance — for a diffusion, the expected integrated local variance to  $T$ . Two pricers:

- **Static replication** (default): the log-contract weights

$$I(T) = 2 \int_0^1 \frac{P(T, x)}{x^2} dx + 2 \int_1^\infty \frac{C(T, x)}{x^2} dx, \quad (5)$$

integrated by trapezoid over the *PDE strike grid* itself (from the lowest quoted strike to  $x_{\max}$ ) — no separate replication grid.

- **Source PDE** (opt-in, `varSwapMethod=source_pde`): solve one *backward* PDE  $\partial_t g + \frac{1}{2}\nu x^2 \partial_{xx} g + \nu = 0$ ,  $g(T, \cdot) = 0$ , and read  $I(T) = g(0, 1)$  — the Feynman–Kac representation of  $\mathbb{E} \int_0^T \nu(t, X_t) dt$ . It reuses the same tridiagonal machinery marched backward, carries *analytic* sensitivities  $\partial I / \partial \theta$  (and  $\partial I / \partial a$  for the free left slope) through the same multi-RHS trick, and has no  $x_{\max}$ -truncation tail. Note 08 develops both in full.

## 6 The calibration objective

The fit minimizes a stacked, bound-constrained least-squares residual over  $\theta$ :

$$\min_{\theta \in [\nu_{lo}, \nu_{hi}]^m} \frac{1}{2} \left( \|r_{\text{opt}}\|^2 + \|r_{\text{vs}}\|^2 + \|r_{\text{bsk}}\|^2 + \lambda \|L(\theta - \theta_{\text{ref}})\|^2 + w_{\text{cvx}} \|r_{\text{cvx}}\|^2 + w_{\text{ft}} \|r_{\text{ft}}\|^2 \right), \quad (6)$$

with blocks:

- **Option residuals**  $r_{\text{opt},q} = (P_q(\theta) - y_q) / \eta_q$  in vega-normalized forward-call units, with  $\eta_q = 0.01 \max(\text{vega}, 10^{-3}) / \sqrt{\omega_q}$ ; or the bid–ask/haireut band hinge with a small mid anchor of weight 0.05 (`midAnchorWeight`) when band edges are present (Note 07).
- **Var-swap residuals**  $r_{\text{vs}} = (I(\theta) - z) / \zeta$  in total variance (Note 08).
- **Operator baskets**  $r_{\text{bsk}}$ : the prior-persistence operators (Note 13) enter as *signed baskets* of option prices — one residual per ATM/RR/BF operator, preserving the risk-reversal and butterfly coupling rather than scattering it into synthetic quotes.
- **Roughness**  $\sqrt{\lambda} L(\theta - \theta_{\text{ref}})$ :  $L$  is the *spacing-aware* second difference (cell-width-normalized divided differences, reducing to  $(1, -2, 1)$  on a uniform grid, exact curvature on a non-uniform one), with time-vs-strike balance `gridRegRho`. This is the smoothness that lets a coarse grid generalize.
- **Convex wing** (opt-in): a one-sided hinge  $\sqrt{w_{\text{cvx}}} \text{relu}(-D^2 \sigma)$  on the nodal local *vols*, confined to vertices at or below the  $5\Delta$  put *and strictly below the deepest observed quote* — the extrapolation tail only (section 9).
- **Front tie** (on by default): a light penalty  $\sqrt{w_{\text{ft}}}(\theta_{0,:} - \theta_{1,:})$  tying the unconstrained  $t = 0$  row to the first data-identified expiry row.

## 7 Sensitivities: tangent and adjoint

Everything the optimizer needs follows from differentiating the stepped system (4). Since  $\nu_\theta$  is *linear* in  $\theta$  (2), so is  $A(\theta)$ , with  $\partial A / \partial \theta_\ell = A[\phi_\ell]$  — the same stencil (3) with  $\nu$  replaced by the basis function  $\phi_\ell$ .

**Proposition 4** (Discrete tangent = same factor, extra right-hand sides). *The sensitivity  $s_\ell^n = \partial U^n / \partial \theta_\ell$  obeys*

$$B^{n+1} s_\ell^{n+1} = s_\ell^n + \Delta T A[\phi_\ell] U^{n+1}, \quad s_\ell^0 = 0, \quad (7)$$

*the discretization of the continuous sensitivity PDE  $\partial_T s_\ell = \frac{1}{2} \nu x^2 \partial_{xx} s_\ell + \frac{1}{2} \phi_\ell x^2 \partial_{xx} c$ . All  $m$  sensitivities share the same tridiagonal factor as the value solve, so the whole Jacobian is one factorization*

plus  $m$  extra back-substitutions per step —  $O(N_T N_x m)$ , and the reason the vectorized march carries the columns as its inner SIMD loop.

*Proof.* Differentiate  $B^{n+1}(\theta)U^{n+1}(\theta) = U^n(\theta)$  in  $\theta_\ell$ :  $B^{n+1}s_\ell^{n+1} + (\partial_{\theta_\ell} B^{n+1})U^{n+1} = s_\ell^n$ , and  $\partial_{\theta_\ell} B^{n+1} = -\Delta T A[\phi_\ell]$  by linearity of  $A$  in  $\theta$ . (Sensitivities against FD: test-locked at  $10^{-6}$ .)  $\square$

**Proposition 5** (Discrete adjoint = one backward sweep). *Let  $J(\theta) = f(HU^{1..N}(\theta))$  be any scalar objective reading the marched solution through observation rows  $H$ . Define adjoint states by the backward recursion*

$$(B^n)^\top \lambda^n = H_n^\top \partial f_n + \lambda^{n+1}, \quad \lambda^{N+1} = 0, \quad (8)$$

where  $H_n^\top \partial f_n$  collects the objective’s direct dependence on step  $n$ . Then

$$\frac{\partial J}{\partial \theta_\ell} = \sum_n \lambda^n^\top \Delta T A[\phi_\ell] U^n, \quad (9)$$

at cost  $O(N_T N_x)$  independent of the parameter count  $m$ .

*Proof.* Form the Lagrangian  $\mathcal{L} = f(HU) - \sum_n \lambda^n^\top (B^n U^n - U^{n-1})$  (boundary terms suppressed). Stationarity in each  $U^n$  gives (8);  $\partial \mathcal{L} / \partial \theta_\ell$  then picks up only the explicit  $\theta$ -dependence of  $B^n$ , which is  $-\Delta T A[\phi_\ell]$ , giving (9). This is summation-by-parts: the transpose of a lower block-bidiagonal system is upper block-bidiagonal, i.e. a backward march with the *transposed* tridiagonal factors.  $\square$

Production uses *both* routes, each where it is optimal. The Gauss–Newton solver needs Jacobian *columns*, so the tangent (7) feeds the multi-RHS march; but the solver is matrix-free — the `LinearizedJacobian` operator exposes exactly two products, the tangent  $Jv$  and the adjoint  $J^\top w$ , and the preconditioned LSMR inner solver touches nothing else. The adjoint product identity  $\langle Jv, w \rangle = \langle v, J^\top w \rangle$  and the gradient itself are test-locked against finite differences.

### Caution

The one production crash in this machinery was an interface bug, not a math bug: `LinearizedJacobian` deliberately has no `.T` attribute (it is an operator, not a matrix), and the early-stop handler, reached only on the rare GN→TRF fallback path that then stalls, called `jac.T @ r` to compute the final gradient — an `AttributeError` in the field (the R1 backtest finding). The fix dispatches to `apply_jacobian_transpose`, and the exact fallback-then-stall path is now test-locked. Lesson: an object that *pretends* to be a matrix must either implement the whole contract or make each unsupported member fail at construction, not mid-fit.

## 8 Solvers and the speed story

The cold fit of (6) was, at  $\sim 533$  vertices, an  $\sim 86$ -second affair hitting the 200-evaluation cap — the heaviest single computation in the app. Driving it down is most of the engineering in this model, and the lessons are worth stating because several plausible ideas *failed*.

### 8.1 Where the time goes

On a cold fit the cost splits roughly as: optimizer linear algebra / dense SVD  $\sim 52\%$ , PDE sensitivity march  $\sim 32\%$ , Jacobian assembly  $\sim 14\%$ , value solve  $\sim 2\%$ . No single per-evaluation lever moves



the total — which is why four separate “faster per evaluation” attempts (coarse grid, first Numba try, Rannacher, and the *first* matrix-free GN) each won only  $\sim 1.1\text{--}1.2\times$  and were shelved.

## 8.2 What worked

### Performance

The cold fit is now  $\sim 3\text{--}6\times$  faster than the banded baseline, recalibrations near-instant. The levers, in order of impact:

1. **Stall-based early-stop.** Track the option-block misfit and stop at the best iterate once it stalls. Fewer evaluations scale march, assembly *and* optimizer together: measured  $1.45\times$  (SPY,  $200\rightarrow 109$  evals) to  $3.3\times$  (NVDA,  $200\rightarrow 41$ ), at  $+0.10\text{--}0.25$  vol bp.
2. **Compiled vectorized-Thomas march.** A `@njit` no-pivot Thomas with the sensitivity columns as the contiguous SIMD inner loop and the source fused into the forward sweep:  $6.1\text{--}6.9\times$  the LAPACK banded march, exact to  $\sim 10^{-15}$ . The first Numba attempt (column-outer scalar Thomas) won only  $1.2\times$  — the lever was the *loop order*, not the compiler.
3. **Matrix-free Gauss–Newton** (default `lvSolver=gn`). LM-damped projected GN with Jacobi column equilibration and matrix-free LSMR on  $Jv/J^T w$  — no dense SVD. First judged non-viable; *promoted to default* once the march became cheap enough that the SVD share dominated:  $1.3\text{--}1.65\times$  over TRF, byte-close surfaces (golden-locked), automatic TRF fallback on breakdown.
4. **Warm seeds.** Previous-surface seed on recalibration ( $\sim 38\times$ ) and a parametric-Dupire cold seed ( $1.3\text{--}1.8\times$ ), both with  $\theta_{\text{ref}}$  held fixed so the regularized optimum is unchanged (byte-identity test-locked).

The order-of-magnitude wins are spent; the remaining levers are incremental.

### Caution

Removing the dense SVD made fits *slower* at  $\leq 440$  vertices, because there the bottleneck was the march, not the SVD — the SVD wall is a  $\gtrsim 1000$ -vertex (future non-tensor “bowtie” grid) phenomenon. The clean synthetic benchmark (zero residual, interior optimum) hid this: GN converged in 8 evaluations there but blew past the cap on stiff real data. Judge speed-ups on real chains; distrust a synthetic-only win. The same benchmark discipline later caught the convex-wing regression (section 9) that a synthetic test could never see.

## 9 Case files

### Case file: the six-day weekly that broke the strike grid

**Setup.** A true 6-DTE SPY weekly (the Massive capture `lv_weekly_massive.json`; the Bloomberg fixture’s shortest expiry is three weeks, which is why the bug survived the benchmark). Normal expiries fit at a few vol bps.

**Failure.** The weekly fit was catastrophic: 108 vol bp RMS against the parametric model’s  $\sim 47$  on the same quotes — the LV surface visibly missing the smile’s curvature.

**Diagnosis — measure first.** A pure side-channel diagnostic (`affine_diag.py`) counts, per expiry: in-range strike vertices (Cause A), vega-floored quotes (Cause B), PDE time steps (Cause C), active prior rows (Cause E). The counts acquitted the usual suspects — time steps adequate, prior inactive, vega floor latent but not binding — and convicted Cause A: the delta strike axis is sized to the *longest* expiry and clipped to the global range, so the weekly’s narrow smile landed *three* vertices on its sharpest curvature (fig. 1).

**Fix.** Two gated changes. (i) A per-expiry coverage floor (`gridXMinPerExpiry=8`): split the widest in-range gaps until every expiry has  $\geq m_{\min}$  vertices inside *its own* traded range — densifying only under-covered short fronts. (ii) A short-expiry-aware PDE step: refine the shared  $\Delta x$  to  $0.3 \times$  the smallest ATM  $\sigma\sqrt{\tau}$ , snapped so  $x=1$  stays a node, clamped to  $[1/400, 0.01]$ .

**Verdict.**  $108 \rightarrow \sim 28$  vol bp from the coverage floor alone (the short-end noise floor), 23.5 with the refined step — now *better* than the parametric fit — while every long expiry and every normal surface is byte-identical (both gates test-locked).

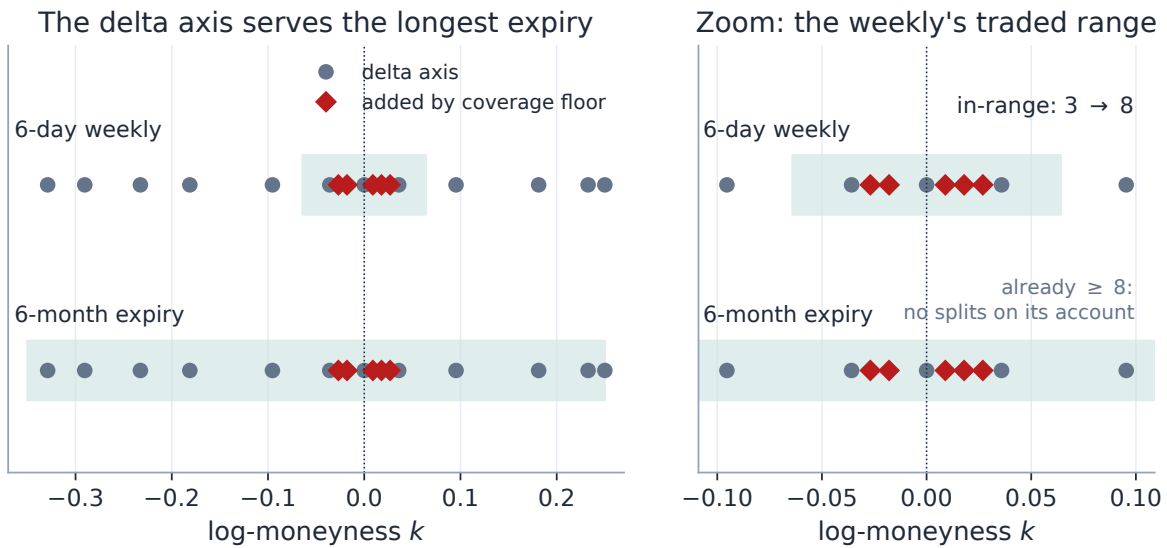


Figure 1: The short-dated rescue, computed by the production grid builder. The delta axis is sized by the longest expiry, so the six-day smile’s traded range (top lane) catches only 3 of its vertices; the coverage floor splits the weekly’s widest in-range gaps until it owns  $\geq 8$ . The six-month expiry already meets the floor, so no split is made on its account — the new vertices all land inside the weekly’s narrow range.

#### Case file: the convex wing that flattened SPY

**Setup.** The convex-wing hinge (section 6) exists to keep the *unquoted* deep-put extrapolation convex. A user ran with a saved `gridXNodes=20` — denser than the default 12.

**Failure.** SPY’s surface RMS degraded to 25.7 vol bp (the “26 bp” regression) on the Bloomberg benchmark; NVDA looked fine.

**Diagnosis.** The constraint originally selected *every* vertex at or below the  $5\Delta$  put, regardless of data. At 20 strike nodes several of those vertices sit *inside* SPY’s densely quoted put wing, and the hinge fought the quotes — forcing the wrong wing on a low-vol name. NVDA’s naturally

convex wing had hidden the bug, and so had the default grid: only the denser grid pushed constrained vertices into quoted territory.

**Fix.** Confine `convex_cols` to vertices at or below the  $5\Delta$  put *and strictly below the deepest observed quote* — the extrapolation tail only.

**Verdict.** SPY 25.7  $\rightarrow$  2.6 vol bp, NVDA unchanged; the confinement is test-locked. The lesson echoes Notes 05 and 09: *a shape constraint must not be imposed where the data already speaks* — the same principle, at the de-Am input, the SIV output, and here the LV extrapolation tail.

## 10 Worked examples

### 10.1 Synthetic round trip

A known skewed local-variance surface is priced through the forward PDE, quotes are generated at four expiries, and the surface is recovered from a *flat* seed by the production calibrator. Figure 2 shows the recovered surface and the implied-vol fit: truth is recovered to a maximum of 0.6 vol bps in 33 evaluations (1.05 s wall, 28 vertices), the skew faithfully reconstructed.

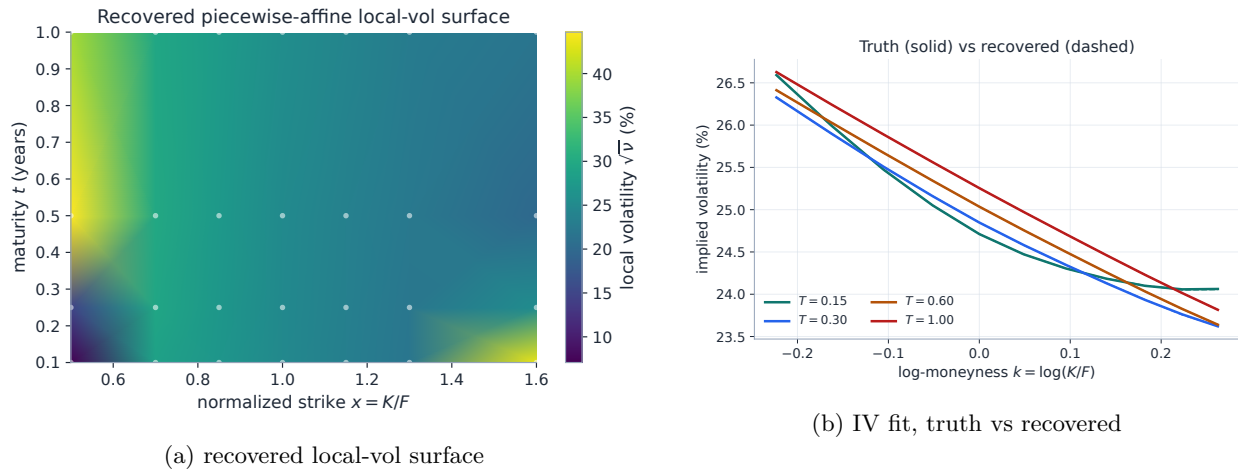


Figure 2: Synthetic round trip through the real calibrator: the skewed surface is recovered to 0.6 vol bps from a flat seed.

### 10.2 The Bloomberg benchmark

On the static Bloomberg fixture the production fit reaches 2.8 vol bps surface RMS on SPY and 11.7 on NVDA (table 1, fig. 3), with no node at a bound on the easy expiries. The harder NVDA short expiries carry the larger residual — the regime the case file’s coverage floor addresses.

Table 1: Production local-vol fit on the Bloomberg fixture (fresh run).

| Name | surface RMS (bp) | worst expiry (bp) |
|------|------------------|-------------------|
| SPY  | 2.8              | 4.4               |
| NVDA | 11.7             | 12.4              |

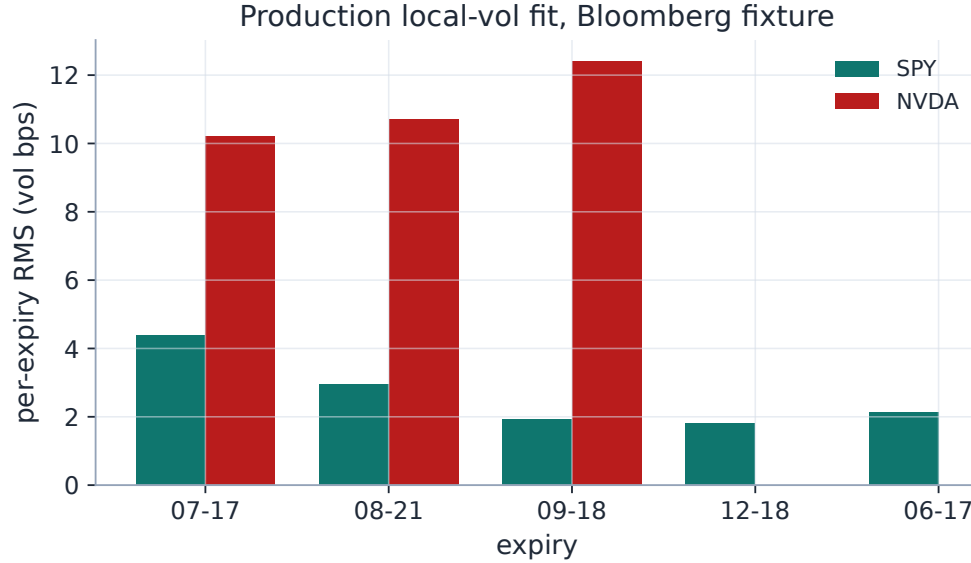


Figure 3: Per-expiry weighted RMS for SPY and NVDA on the Bloomberg fixture. The short end is where the residual lives — and what the coverage floor and adaptive PDE step protect.

## 11 What is original, and limitations

The  $P_1$ -positivity guarantee and forward-calibration philosophy are Andreasen–Huge; the contributions here are depth and discipline. The *vectorized-Thomas Numba march* (loop order as the lever,  $6.5\times$ ); the *stall early-stop* that scales the whole fit; the *matrix-free GN* first rejected, then correctly promoted when the march made the SVD dominant — with the tangent/adjoint pair of section 7 as its only interface to the PDE; the *source-PDE var-swap* with analytic sensitivities; and the *measure-first* short-dated diagnosis that convicted the strike grid on counts rather than hunches. Each was validated against a real benchmark that twice caught regressions a synthetic test would have missed. Limitations: a pure diffusion cannot make a true short-dated event spike (Note 11’s variance clock absorbs scheduled events instead); the tensor grid wastes vertices in the corners (the future non-tensor bowtie grid is where the GN no-SVD advantage will finally dominate); and the fit, though much faster, remains the heaviest computation in the app.

## 12 Traceability

Table 2: Claims in this note and the code/tests that lock them.

| Claim                                                                            | Object | Code anchor                            | Test anchor                                                                                                                          |
|----------------------------------------------------------------------------------|--------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| $P_1$ basis: partition of unity; proposition 2<br>nodal bounds bound the surface |        | <code>models/localvol/affine.py</code> | <code>test_localvol_affine.py::test_basis_partition_of_unity_and_nodal_interpolation,::test_nodal_bounds_imply_surface_bounds</code> |

| Claim                                                                       | Object                           | Code anchor                          | Test anchor                                                                                                                                                   |
|-----------------------------------------------------------------------------|----------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calibrated surface is arbitrage-<br>clean; density has no interior<br>zeros | proposition 3<br>and corollary 1 | models/localvol/affine.<br>py        | test_localvol_<br>affine.py::test_<br>calibrated_surface_<br>is_arbitrage_free,<br>test_api_affine.py::<br>test_affine_density_is_<br>clean_no_interior_zeros |
| Scheme converges (flat + skew<br>round trips)                               | section 4                        | models/localvol/pde.py               | test_localvol.py::test_<br>spatial_convergence,<br>::test_skew_dupire_<br>round_trip                                                                          |
| Tangent sensitivities exact vs<br>FD                                        | proposition 4                    | models/localvol/affine.<br>py        | test_localvol_affine.<br>py::test_forward_<br>sensitivities_match_<br>finite_differences                                                                      |
| Adjoint product and gradient<br>correct                                     | proposition 5                    | models/localvol/affine_<br>gn.py     | test_affine_gn.py::<br>test_jacobian_transpose_<br>inner_product_identity,<br>::test_gradient_alpha_<br>test                                                  |
| GN reaches the TRF optimum;<br>fallback-then-stall safe (R1)                | sections 7 and 8                 | models/localvol/affine_<br>gn.py     | test_affine_gn.py::<br>test_gn_matches_trf_on_<br>golden, ::test_gn_trf_<br>fallback_then_stall_<br>returns_surface                                           |
| Numba march = banded march                                                  | listing 1                        | models/localvol/affine_<br>march.py  | test_affine_march.py::<br>test_numba_march_<br>matches_banded_<br>prices_and_sens, ::<br>test_numba_calibration_<br>matches_banded_surface                    |
| Convex wing confined to the<br>extrapolation tail                           | section 9                        | api/affine_fit.py<br>(_resolve_grid) | test_affine_grid_design.<br>py::test_convex_wing_<br>confined_to_quoted_<br>extrapolation                                                                     |
| Coverage floor / PDE step<br>touch only short fronts                        | section 9                        | api/affine_fit.py                    | test_affine_grid_design.<br>py::test_coverage_<br>floor_densifies_<br>only_the_short_front,<br>::test_pde_dx_refines_<br>only_short_surfaces                  |
| Warm start leaves the opti-<br>mum unchanged                                | section 8                        | api/affine_fit.py                    | test_affine_warm_start.<br>py::test_recalibration_<br>warm_starts_and_cuts_<br>evals                                                                          |
| Source-PDE var swap + free<br>left slope                                    | section 5                        | models/localvol/varswap_<br>pde.py   | test_varswap_source.py,<br>test_affine_grid_design.<br>py::test_free_a_reduces_<br>varswap_error                                                              |

| Claim                                      | Object     | Code anchor                           | Test anchor                                                                                                        |
|--------------------------------------------|------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Diagnostics are a pure side-channel        | section 9  | api/affine_diag.py                    | test_affine_diag.py::<br>test_record_shape_and_purity                                                              |
| Dupire extraction returns NaN on arbitrage | appendix C | models/localvol/dupire.py             | test_localvol.py::<br>test_dupire_arbitrage_returns_nan                                                            |
| Prior operators enter as signed baskets    | section 6  | api/affine_fit.py,<br>affine_calib.py | test_affine_basket.py::<br>test_basket_pulls_<br>surface_toward_target,<br>::test_empty_baskets_<br>byte_identical |

## A Hyperparameter atlas

Table 3: Local-volatility hyperparameters.

| Knob                                       | Default            | Role                                                                                                  |
|--------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------|
| <i>Surfaced (OptionsSettings)</i>          |                    |                                                                                                       |
| gridStrikeMode                             | delta              | Strike-vertex placement (delta-spaced).                                                               |
| gridXNodes                                 | 12                 | Strike vertices of the variance surface.                                                              |
| gridXMinPerExpiry                          | 8                  | Min in-range strike vertices per expiry (section 9).                                                  |
| gridTNodes                                 | 10                 | Maturity vertices.                                                                                    |
| gridRegLambda $\lambda$                    | $10^{-2}$          | Roughness strength (model-layer signature default is $10^{-4}$ ; production passes this).             |
| gridRegRho $\rho$                          | 1.0                | Time-vs-strike roughness balance.                                                                     |
| convexWing                                 | / false / $10^3$   | Convex-wing hinge (extrapolation tail only).                                                          |
| convexWingWeight                           |                    |                                                                                                       |
| frontTie                                   | / true / $10^{-2}$ | Front-end tie of the $t = 0$ row to the first expiry row.                                             |
| frontTieWeight                             |                    |                                                                                                       |
| lvVolCapMult                               | 3.0                | Adaptive variance cap $\min(\max(0.36, (\text{mult} \cdot \sigma_{\max})^2), 16)$ — at most 400% vol. |
| leftWingSlopeMult                          | 1.5                | Left local-variance extrapolation slope multiple.                                                     |
| varSwapMethod                              | static             | Var-swap pricer: replication over the PDE grid, or source-PDE.                                        |
| lvSolver                                   | gn                 | Matrix-free Gauss–Newton (default) or trf.                                                            |
| lvFastKernel                               | true               | Numba vectorized-Thomas march.                                                                        |
| lvEarlyStop                                | true               | Stall-based early stop.                                                                               |
| timeScheme                                 | implicit           | Implicit Euler (monotone) or opt-in Rannacher/CN (remark 1).                                          |
| midAnchorWeight                            | 0.05               | Mid anchor inside the band objective.                                                                 |
| <i>Hidden (affine PDE / grid / solver)</i> |                    |                                                                                                       |

| Knob                                                               | Default                 | Role                                                                                   |
|--------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------|
| varLo / varHi                                                      | 0.0025 / 0.36           | Request-level nodal variance box (vol 5%/60%) before the adaptive cap.                 |
| _X_DX / _X_MAX_MIN                                                 | 0.01 / 2.5              | PDE strike step and minimum $x_{\max}$ .                                               |
| _DT_MAX                                                            | 0.01                    | Implicit-Euler maturity step (0.03 under Rannacher).                                   |
| rannacher_steps                                                    | 2                       | Implicit start-up steps before CN (opt-in scheme).                                     |
| _PDE_DX_SHORT_FRAC / _PDE_N_MAX                                    | 0.3 / 400               | Short-dated step: $0.3\sigma\sqrt{\tau}$ , clamped $[1/400, 0.01]$ .                   |
| _DELTA_SET                                                         | 7 deltas                | Put deltas $\{1, 2, 5, 10, 25, 40, 50\}\%$ (and call mirrors) spacing the strike axis. |
| _CONVEX_WING_DELTA                                                 | 0.05                    | Wing boundary for the convex hinge ( $5\Delta$ put).                                   |
| _VOL_TOL / _VEGA_FLOOR                                             | 0.01 / $10^{-3}$        | Vega normalization of option residuals.                                                |
| x_scale / tolerances                                               | jac / $10^{-8}$         | Optimizer scaling; <code>max_nfev</code> = 200.                                        |
| _STALL_WINDOW/_RTOL                                                | 12 / $5 \times 10^{-3}$ | TRF early-stop ( $18 / 3 \times 10^{-3}$ under GN).                                    |
| VAR_FLOOR                                                          | $10^{-6}$               | Dupire-extraction local-variance floor.                                                |
| <i>Hidden (separate log-strike validator, <code>pde.py</code>)</i> |                         |                                                                                        |
| DEFAULT_N_K / DEFAULT_DT_MAX                                       | 1201 / 1/400            | Validator grid.                                                                        |
| N_RANNACHER                                                        | 4                       | Validator Rannacher quarter-steps.                                                     |
| SPAN_SD / SPAN_MIN                                                 | 7.5 / 0.6               | Validator grid span (sd / floor).                                                      |

## B Performance notes

1. **Numba march** (`lvFastKernel`). Factor-once no-pivot Thomas (proposition 3 is why no pivoting),  $m$  sensitivity columns as the contiguous inner SIMD loop, fused source:  $6.1\text{--}6.9\times$  vs LAPACK banded, exact to  $\sim 10^{-15}$ ; graceful banded fallback without Numba.
2. **Early stop** (`lvEarlyStop`). Best-iterate stop on option-block stall:  $1.45\times$  (SPY) to  $3.3\times$  (NVDA) at  $+0.10\text{--}0.25$  vol bp; window 0 is byte-identical.
3. **Matrix-free GN** (`lvSolver=gn`). Preconditioned LSMR on the tangent/adjoint products of section 7; no dense SVD.  $1.3\text{--}1.65\times$  over TRF at tensor sizes; golden-locked to the TRF optimum; automatic TRF fallback; sparse CSR regularization block.
4. **Warm starts**. Previous-surface recalibration  $\sim 38\times$ ; parametric-Dupire cold seed  $1.3\text{--}1.8\times$ ;  $\theta_{\text{ref}}$  fixed so the optimum is unchanged.
5. **Shelved (documented in the perf roadmap)**: coarse-grid calibration (biases  $\theta$  by up to  $\sim 26$  vol points — non-viable), first Numba attempt (wrong loop order,  $1.2\times$ ), Rannacher/CN ( $\sim 1.1\times$  and a measured coarse- $x$  arbitrage — remark 1), GN for the non-smooth band objective, thread/process parallelism (GIL-negative). Five dead-ends confirming the cost is distributed, not localized.

## C Reference implementation: Dupire extraction

Forward calibration never differentiates the data (section 2), but the *inverse* map — reading a local-variance grid off an already-fitted total-variance surface  $w(k, T)$  — is Gatheral’s formula [3], used for grid export, the parametric cold seed of section 8, and the graph LV projection (Note 14). The denominator  $g$  is the butterfly function:  $g \leq 0$  marks strike arbitrage in the implied surface, where no positive local variance can reproduce it — returned as `nan`, never clipped. Matches production to  $10^{-13}$ .

Listing 2: Dupire/Gatheral local variance from an implied total-variance surface (distilled from `models/localvol/dupire.py`).

```
1 def dupire_local_variance(k, w, wk, wkk, wt):
2     r = k/w                                     # w = w(k, T), with k/kk/T
3     derius
4     g = 1 - r*wk + 0.25*(-0.25 - 1/w + r*r)*wk*wk + 0.5*wkk
5     return np.where((w > 0) & (g > 0),      # g <= 0: butterfly arbitrage
6         -> nan
7         wt/np.where(g > 0, g, 1.0), np.nan)
```

## References

- [1] B. Dupire. Pricing with a smile. *Risk*, 7(1):18–20, 1994.
- [2] J. Andreasen and B. Huge. Volatility interpolation. *Risk*, March 2011.
- [3] J. Gatheral. *The Volatility Surface*. Wiley, 2006.
- [4] M. Giles and R. Carter. Convergence analysis of Crank–Nicolson and Rannacher time-marching. *J. Comput. Finance*, 9(4), 2006.
- [5] A. Berman and R. Plemmons. *Nonnegative Matrices in the Mathematical Sciences*. SIAM, 1994.